

Earth and Planetary System



Introduction to Geophysics



Prof. A.K. Chaubey
akchaubey@iitism.ac.in

* Extra Class

L1: 05 Jan. 2024
L2: 08 Jan. 2024
L3: 10 Jan. 2024
L4: 12 Jan. 2024
L5: 12 Jan. 2024*
L6: 15 Jan. 2024
L7: 17 Jan 2024

SEMESTER - IV

Course Type	Course Code	Name of Course	L	T	P	Credit
ESO3	GPE 201	Earth and Planetary System	3	0	0	9

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Course Objective

The primary objective of the course is to introduce fundamental aspects of Earth and Planetary system and its related changes with time. This course will emphasize the knowledge on the branches of geophysics, solar system, planets, climates, ocean, carbon cycle, and transitions of Earth's structure through different geological ages. The underlying physics of the various Earth and planets related changes and its creation are presented through theory classes.

Learning Outcomes

Knowledge on the history of Earth's development, the solar system; climate and its changes; Earth's interior; plate tectonics; the physical property of ocean water; changes of ocean climates; global carbon cycle; depositional events and environments with geological ages and glaciations are the goal of the course.

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Introduction to Geophysics, different branches of Geophysics and relationship with other sciences.	4	Presentation of geophysics and its branches
2.	Solar System formation, meteorites, planet formation, Asteroid belt, Nebula Hypothesis, Kepler's Law, Bode's law	4	Knowledge of the solar system
3.	Isotopes and Minerals	2	Fundamentals of minerals
4.	Geomorphology and Geophysical Signature observed by various missions	3	Study of comparative planetology

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
5.	Plate tectonics, physical geology systems explaining the interior of the Earth	6	Study on Earth's tectonic system
6.	Blackbody radiation, energy balance, greenhouse effect, climate forcings, climate sensitivity, the role of sun/volcanoes/greenhouse gasses/aerosols, climate feedbacks	4	Overview and fundamental concept on Earth's climates system
7.	Structure and circulation of the atmosphere, Coriolis effect, Geostrophic balance, Wind-driven circulation, Thermo-Haline circulation, Upwelling, El Nino-Southern Oscillation, Monsoons	5	Knowledge of ocean-atmospheric circulation

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
8.	Carbon reservoir and fluxes, long-term carbon cycle and plate tectonics, Volcanic outgassing and silicate weathering, glacial-interglacial CO ₂ cycles, and role of the Ocean, the anthropogenic perturbation, Keeling curve, Carbon uptake by Ocean and the terrestrial Biosphere	4	Fundamental study on the global carbon cycle
9.	Major climate events and trends during the Cenozoic last 65 Million	4	Knowledge of climate changes during Cenozoic
10.	Late Paleocene Thermal Maximum, Eocene climate optimum, Antarctic glaciation, Northern Hemisphere glaciation, Plio-Pleistocene Cooling	3	Glaciation and physical property changes of Earth during different geological age
	Total Class	39	

Text Books

- Fowler, C.M.R., Solid Earth: An Introduction to Global Geophysics
- Howell, B.F., An Introduction to Geophysics, Mc-Graw Hill
- Lowrie, W., Fundamentals of Geophysics, Cambridge University Press

Reference Books

- Jacobs, J.A., A Text Book of Geonomy, Adam-Hilger
- Tucker, R.H., Cook, A.H., Iyer, H.M. and Stacey, F.D., Global Geophysics, English Univ. Press
- Donald, L., Turcotte & Gerald Schubert : Geodynamics (Second Edition)

Lecture timings

Day	Time	Venue
Monday	08:00 – 08:50	NLHC-G8
Wednesday	08:00 – 08:50	NLHC-G8
Friday	08:00 – 08:50	NLHC-G8

Examinations during the Semester

1. Quiz-01: Total Questions {Short answer type (~70%) and numerical (~30%)} : 10,
Time Duration: 30 minutes, Total Marks: 10
2. Mid-Term Exam: Total Questions {Descriptive answer type (~70%) and numerical (~30%)} : 5,
Time Duration: 2 hours, Total Marks: 50 (32)
3. Quiz-02: Total Questions {Short answer type (~70%) and numerical (~30%)} : 10, Time Duration: 30 minutes, Total Marks: 10
4. End-Term Exam: Total Questions {Descriptive answer type (~70%) and numerical (~30%)} :5,
Time Duration: 3 hours, Total Marks: 100 (48)
5. Grading: As per institute policy.

All Examinations will be offline mode only.

Class Attendance

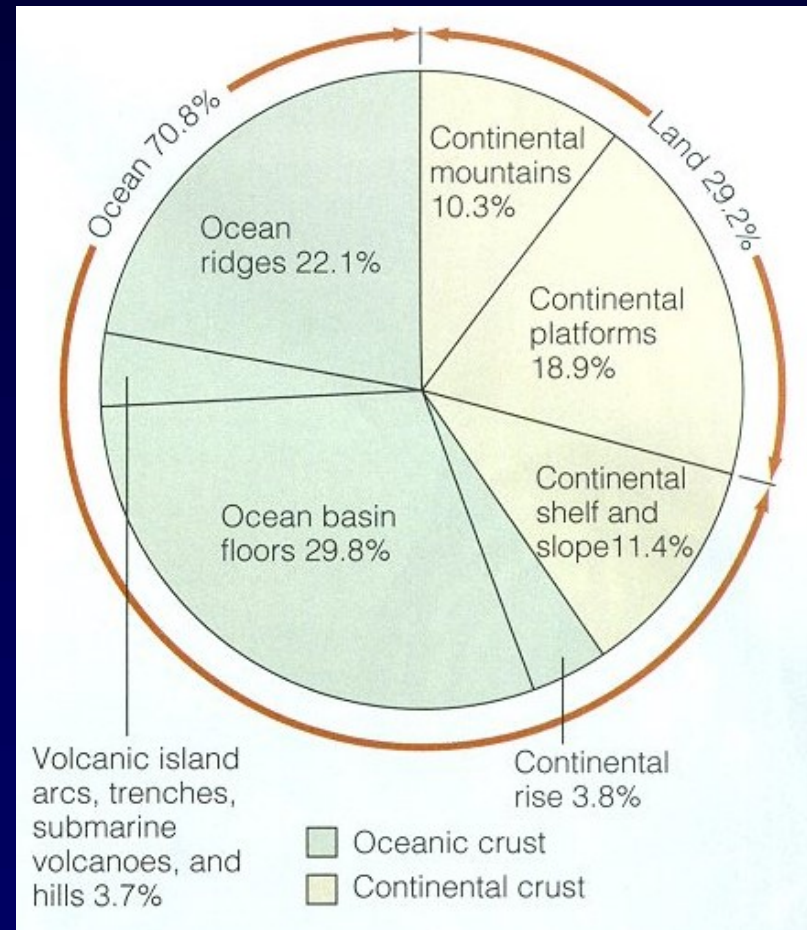
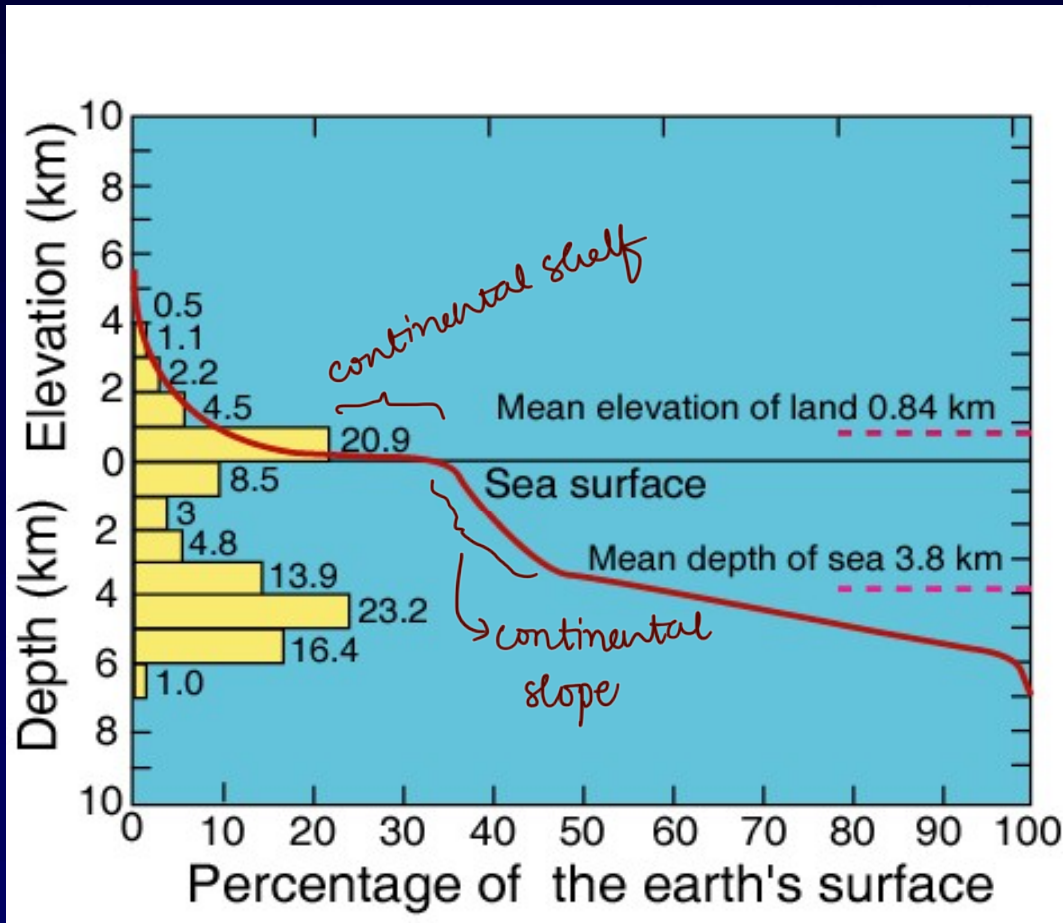
➤ 75% out of total of 39 classes.

What is a Planetary System?

- A planetary system is a group of planets and other bodies circling a star or star system.
- Generally speaking, planetary systems describe systems with one or more planets, although such systems may also consist of bodies such as dwarf planets, asteroids, natural satellites, meteoroids, comets and planetesimals.
- The Sun together with its planetary system, which includes Earth, is known as the Solar System.

The Dynamic Earth in the Solar System

Continent and Ocean



- Substantial part of continental crust lies beneath the oceans.
- Oceans cover an area of 361 million km², and contain a volume of about 1.37 billion km³ of water.
- Northern Hemisphere is covered by 60.7% oceans and 39.3% land, while the Southern Hemisphere is covered by 80.9% oceans and 19.1% land.

Area of Continents and Oceans and Mean Ocean Depths

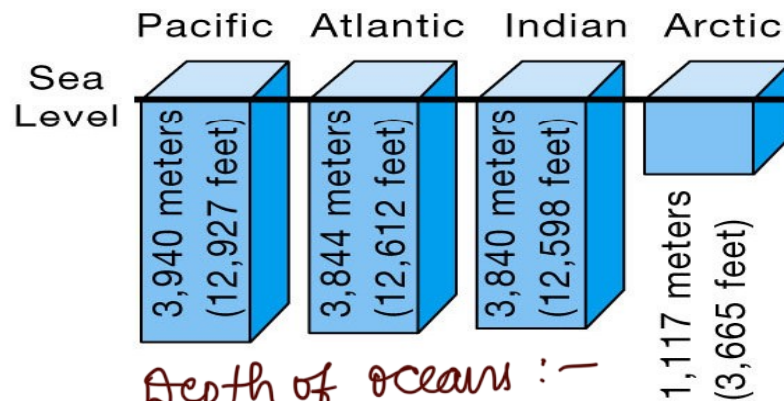
	Area (10 ⁶ km ²)	Percent of land or ocean	Percent of world surface	Mean depth (km)
World surface	510	—	100	—
All continents	148	100	29.2	—
All oceans	362.0	100	70.8	3.729
Eurasia	54.8	36.8	10.8	—
Asia	44.8	29.8	8.7	—
Europe	10.4	7.0	2.1	—
Africa	30.6	20.5	6.0	—
North America	22.0	14.8	4.3	—
South America	17.9	12.0	3.5	—
Antarctica	15.6	10.5	3.1	—
Australia	7.8	5.2	1.5	—
Pacific Ocean	166.2	—	—	4.188
—with adjacent seas	181.3	50.1	35.4	3.940
Atlantic Ocean	86.6	—	—	3.736
—with adjacent seas	94.3	26.0	18.4	3.575
Indian Ocean	73.4	—	—	3.872
—with adjacent seas	74.1	20.5	14.5	3.840
Arctic Ocean	9.5	—	—	1.330
—with adjacent seas	12.3	3.4	2.4	1.117

Data for oceans are from Menard and Smith (1966).

Data for continents are from Ronov and Yaroshevsky (1969).

Oceans and Land

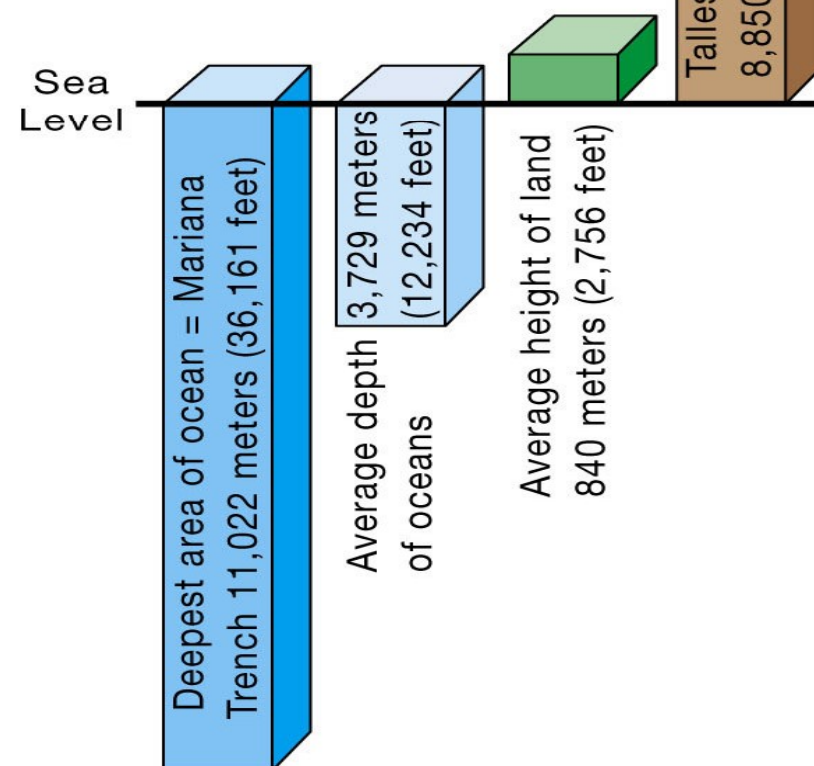
(c) Average Ocean Depth



Depth of oceans :-

- ① Pacific ocean
- ② Atlantic ocean
- ③ Indian ocean
- ④ Southern ocean
- ⑤ Arctic ocean

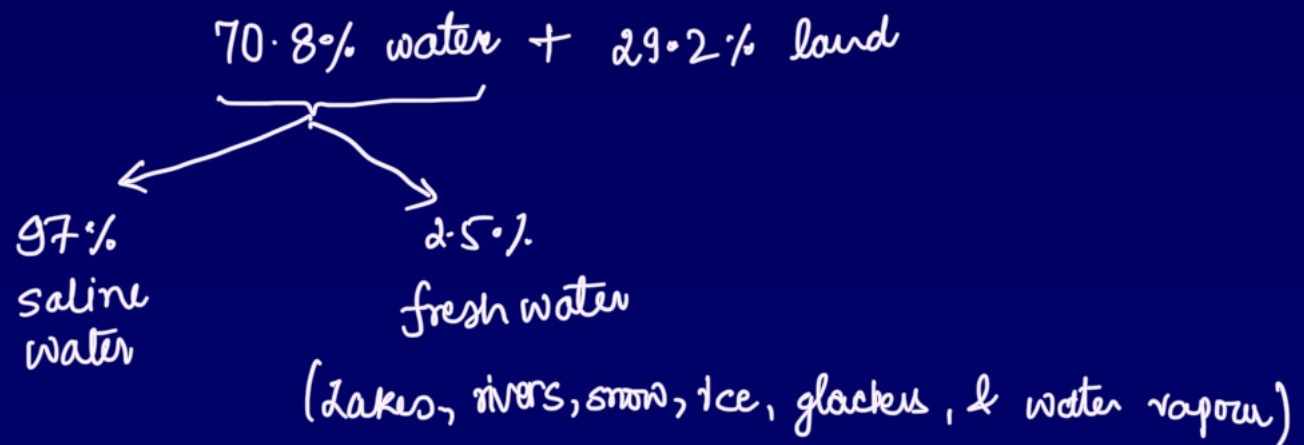
(d) Comparing Oceans to land



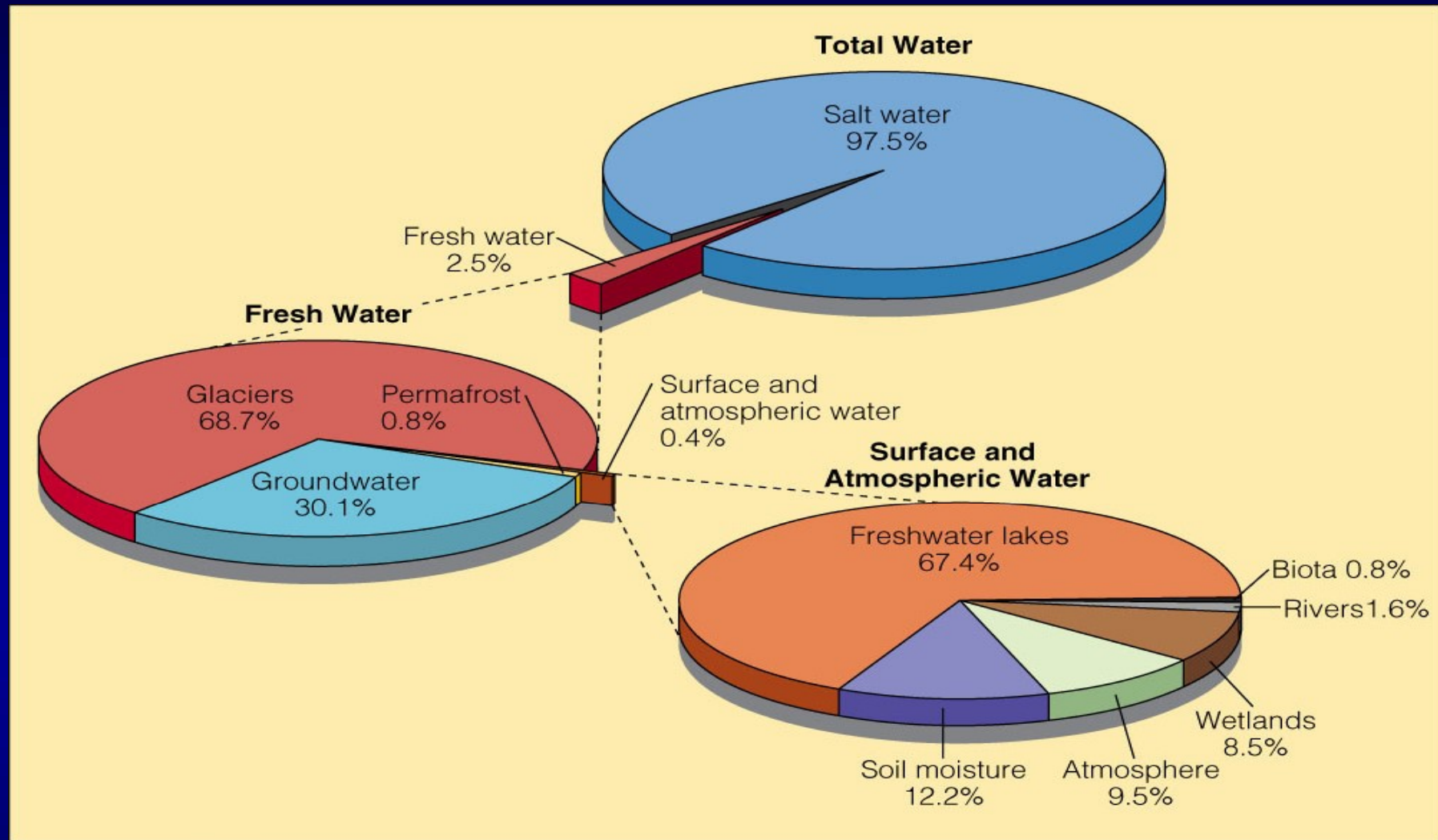
PLANET WATER

70.8% OF EARTH IS COVERED BY WATER

- 97.5% is saline water (oceans and seas)
- 2.5% is fresh water (Lakes, rivers, snow, ice, glaciers, and water vapor in atmosphere)



Summary of the relative amount of water on or near Earth's surface



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More than 97% of the water lies in the ocean. Of all water at Earth's surface, ice on land contains about 1.7%, groundwater 0.8%, rivers and lakes 0.007%, and the atmosphere 0.001%.

NONE, as small as they may be, are unimportant!

The **Pacific Ocean** is the largest ocean. It covers 46% of surface of the total extent of the Oceans and sea of the globe. Average depth is ~3940 m.

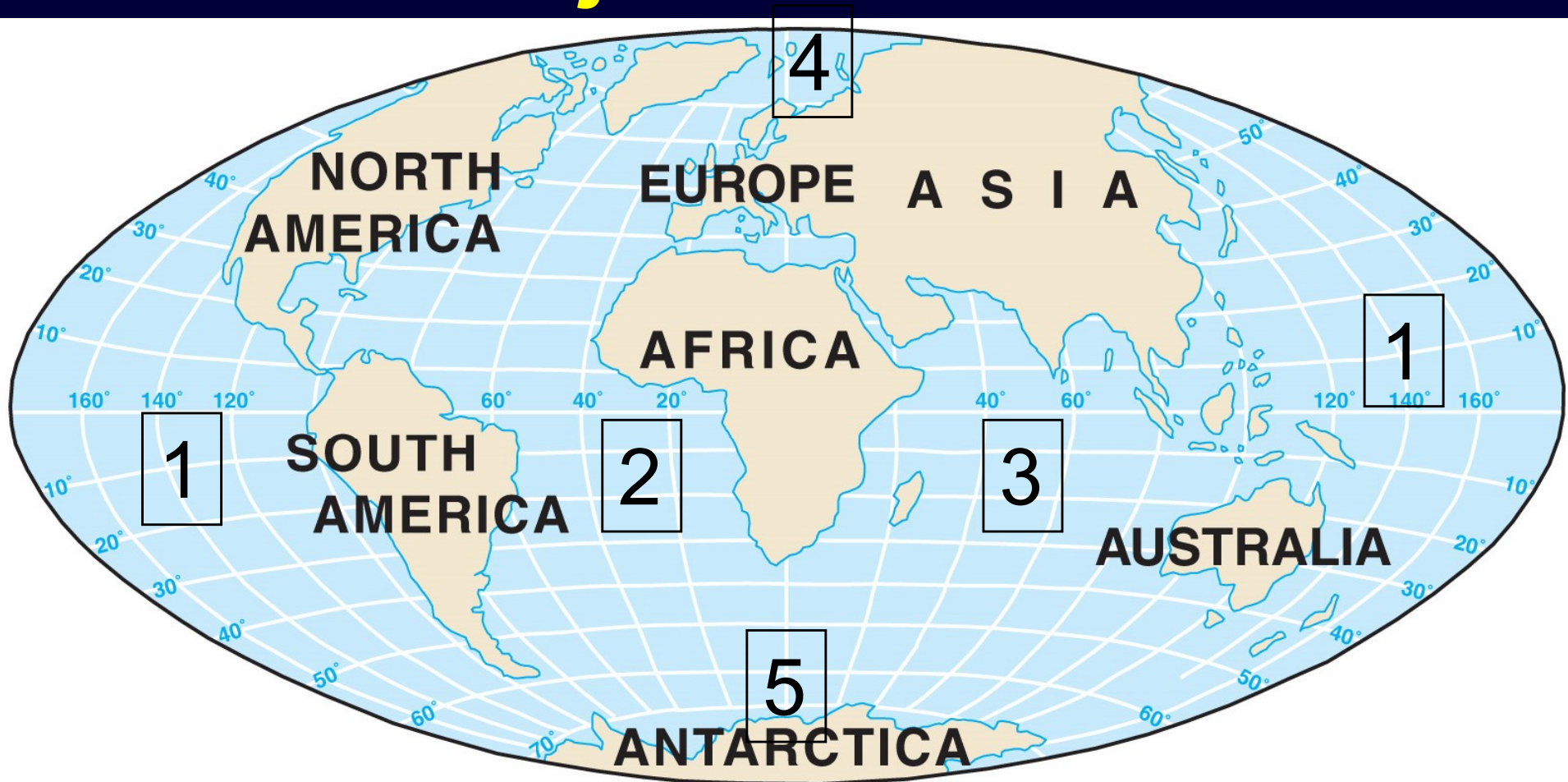
The **Atlantic Ocean** is the second largest ocean. Average depth is ~3575 m.

The **Indian Ocean** is the third largest ocean. Average depth is ~3840 m.

The **Arctic Ocean** is the smallest of the five oceans. The Arctic Ocean is partially covered with ice, the extent of which varies according to the season. Average depth is ~1117 m.

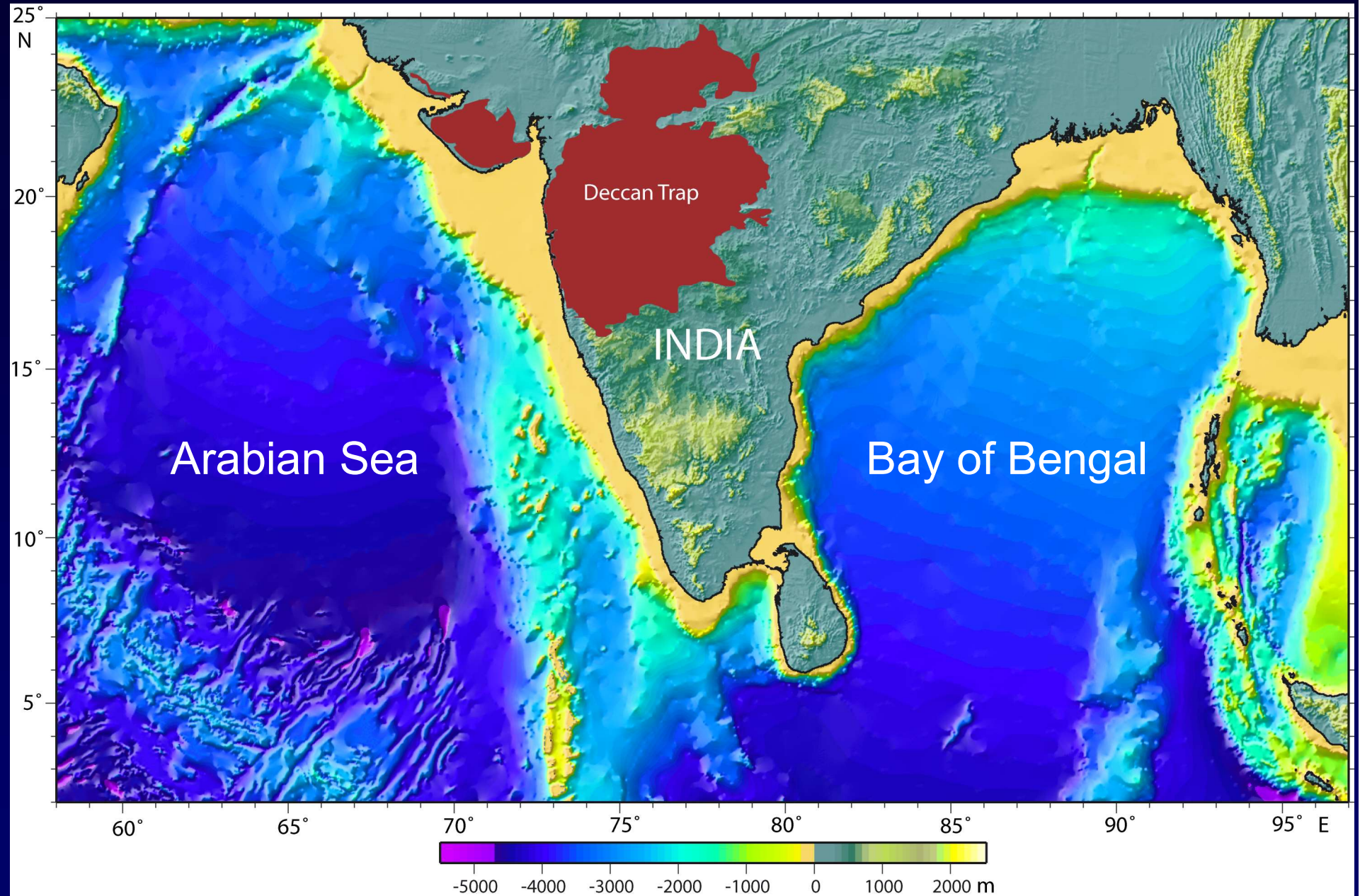
The **Southern Ocean** is the second smallest ocean surrounding the Antarctica. The Southern Ocean is partially covered with ice, the extent of which varies according to the season. Average depth is ~3270 m.

Major Oceans



1. Pacific Ocean, 2. Atlantic Ocean, 3. Indian Ocean, 4. Arctic Ocean and 5. Antarctic/Southern Ocean

Northwest and northeast Indian Ocean



Earth System

1. Atmosphere: The gases that envelop the Earth.
2. Hydrosphere: Water on or near the Earth's surface (rivers, oceans, glaciers, lakes.
3. Biosphere: All living or once-living materials.
4. Geosphere: The solid rocky Earth.

Lithosphere

Earth Heat Engine

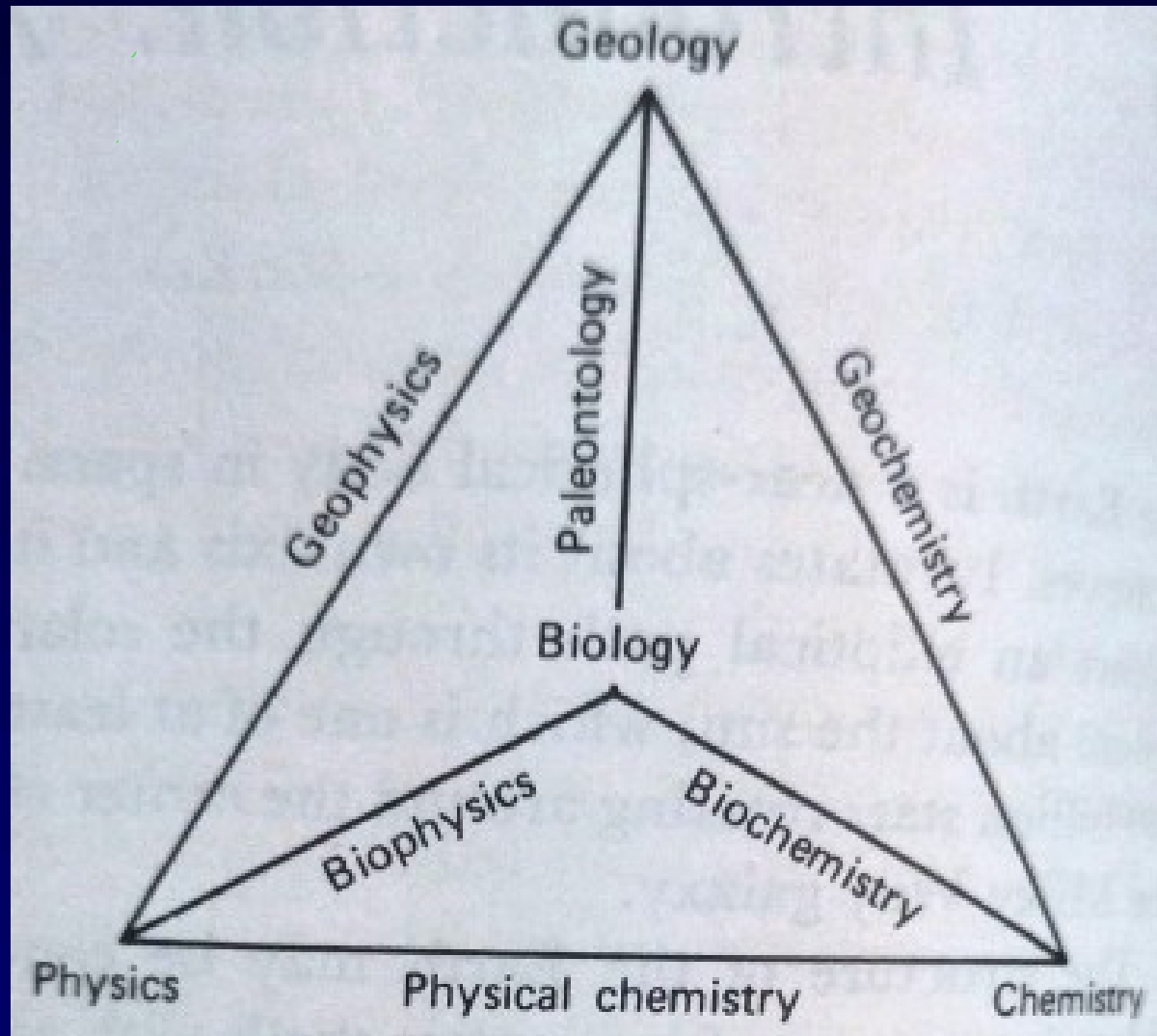
1. External (energy from the Sun): Drives weathering & water cycle phenomena.
 - Primary driver of atmospheric (weather) and hydrospheric circulation
 - Control weathering of rocks at the Earth's surface.
2. Internal (heat moving from hot interior to cooler exterior) (Geospheric phenomena)
 - Primary driver of most geospheric phenomena (volcanism, magmatism, tectonism).

What is Geophysics?

- Every aspect of human life is directly affected by the earth on which we live.
- The air we breathe, the water we drink, the food we eat, the shelter we construct, and the energy we consume are all directly related to man's optimal use and improved understanding of our planet Earth.
- "Geophysics" is that part of observational and experimental physics which pertains to this same planet, particularly its atmosphere, hydrosphere, crust, mantle, and core.

*observational & experimental physics of the earth's atmosphere,
hydrosphere, crust, mantle & core.*

Tetrahedron of Physical Sciences



Tetrahedron of Physical Sciences, which include all of the subjects applied to the study of the Earth.

What is Geophysics?

no use of any instrument for investigation

- Geophysics is non-invasive investigation of subsurface conditions in the Earth through measuring, analyzing and interpreting physical fields at the surface.
- Some studies are used to determine what is directly below the surface or few meters below the surface; other investigations extend to depths of 1000's of meters or more.

Geophysics

- Geophysics is the study of the earth, its surface, interior, and surrounding environment in space using principles of the physics.
- It uses non destructive methods of investigation.
- The study of the Earth's physical properties and of the physical processes acting upon, above and within the Earth.

- Knowledge about structure & composition of Earth's uppermost part comes from study of rocks exposed on surface -- (*the direct techniques*)
- Study of drill holes samples allows extending such knowledge to deeper regions to some extent -- (*also a direct technique*)
- How to know, what exists beyond the reach of drill holes ?

- **Knowledge about the deep subsurface zones of the Earth comes from the *indirect techniques* of geophysical methods.**

Geophysical methods - the underlying foundation

- Like surface, the subsurface of the Earth is also composed of different rocks & minerals
- Tectonic processes & assemblages of rocks, sediments and minerals creates lateral / vertical discontinuities.

Geophysical methods – the underlying foundation ... contd.

- **Different rocks & minerals possess contrasting physical properties & produce measurable effects**
- **At discontinuities, the collective contribution of measurable effects manifests as anomalies**

Geophysics – the basic approach

- **Detect anomalies – *i.e. locate departure from regularity***
- **Deduce causes – *i.e. the location & nature of the features causing the anomalies***

Geophysics - the essence

- **Study of the Earth its surroundings using principles of Physics**
 - **General – Discontinuities in the Earth as a whole**
 - **Exploration – Investigation of the local discontinuities**

Geophysical Exploration Methods

- 1. Passive Geophysical Methods (Natural Field Methods):** Passive geophysical methods incorporate measurements of naturally occurring fields or properties of the earth.

Generally, natural field methods can provide information on Earth properties to significantly greater depths and are logistically more simple to carry out.

Geophysical Methods

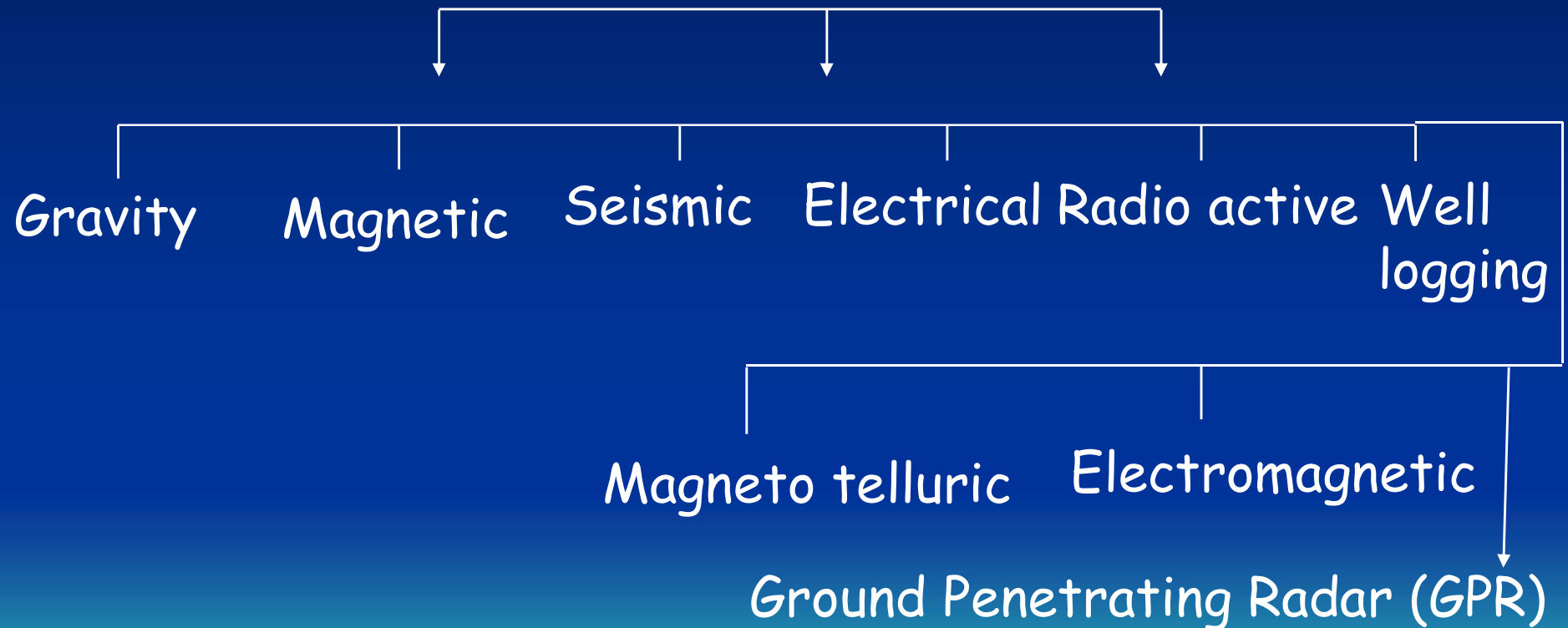
2. Active Geophysical Methods (Artificial Source Methods): In active geophysical methods, on the other hand, a signal is injected into the earth and we then measure how the earth responds to this signal.

These methods are capable of producing a more detailed and better resolved picture of the subsurface geology.

Geophysical Surveys at different elevations

- 1. Land based geophysical surveys**
- 2. Drone-based geophysical surveys**
- 3. Ship-borne geophysical surveys**
- 4. Deep-towed geophysical surveys**
- 5. Geophysical surveys at seabed**
- 6. Airborne geophysical surveys**
- 7. Satellite system for geophysical exploration**

Geophysical Methods



Geophysical Methods and their Application

Gravity and Magnetic Methods

Method	Measured parameter	Physical property	Major applications
Potentials			
Gravity	Spatial variations in the local strength of the gravitational field of the Earth	Local variations in density	Mapping of regional structures, sedimentary basins, salt diapers, plutonic intrusions delineation, sand and gravel deposits, depth to bedrock
Magnetic	Spatial variations in the local strength of the geomagnetic field	Local variations in susceptibility and remanence	Mapping of regional structures, airborne surveys, igneous intrusions, sea floor spreading, salt structures, mineral deposits, buried environmental hazards, archeology

Seismic Methods

Refraction	Traveltimes, amplitudes, waveforms of refracted elastic waves	Compressional, shear wave velocities	Crustal scale to engineering scale mapping of rock types, structural boundaries, foundations, hydrogeology
Reflection	Traveltimes, amplitudes, waveforms of reflected elastic waves	Compressional, shear wave contrasts, density contrasts, seismic image	Oil and gas exploration, site surveying, bedrock mapping, detection of shallow faults and cavities.

Electrical Methods

Electrical			
Resistivity	Earth resistance (applied voltage / measured current)	Electrical resistivity (conductivity)	Mineral prospecting, engineering and hydrogeology, contaminant mapping, construction site investigation, groundwater
Induced polarization (IP)	Voltage decay, or frequency dependent resistance	Electrical capacitance	Detection of disseminated mineral deposits, aquifer mapping, contaminant mapping
Self-potential (SP)	Natural electric potential	Electro-chemical activity	Mineral prospecting, graphite detection, hydrogeology (seepage), geothermal studies
Electromagnetic (EM)	Secondary (induced) electromagnetic fields	Electrical conductivity and inductance	Deep mineral prospecting, airborne surveys, conducting faults, groundwater studies, detection of underground pipes and cables, agricultural studies

Radiometric Method: The radiometric, or gamma-ray spectrometric method is a geophysical method used to estimate concentrations of the radio elements i.e. potassium, uranium and thorium in the near surface. This is carried out by measuring the gamma-rays which the radioactive isotopes of these elements emit during radioactive decay.

Geophysical Borehole Logging Method: Borehole logging systems are used to make measurements inside boreholes (or drill holes) with borehole probes. The probes contain sensors and instrumentation, which transmit the data via a cable, to recording instruments on the surface. A borehole log is a record or graph of the sensor output (e.g., gamma ray count rate) plotted on the horizontal axis versus depth on the vertical axis

Integrated Geophysical Method: Integrated geophysical methods (combinations of two or more methods) are commonly used in the mapping of subsurface geologic discontinuities.

Application Geophysical Methods

Exploration of	Methods
Fossil fuels (oil, gas, coal)	S, G, M, (EM)
Metalliferous mineral deposits	M, EM, E, SP, IP, R
Bulk mineral deposits (sand and gravel)	S, (E), (G)
Underground Utilities	S, M, GPR
Ground Water	E, S, (G), (M), (GPR)
Engineering/construction site investigation	E, S, GPR, (G), (M)
Archaeological investigations	GPR, E, EM, M, (S)
Home Land Defence	S, GPR, M, R

G → Gravity; M → Magnetic; S → Seismic; E → Electrical resistivity;
 SP → Self-potential; IP → Induced polarization; EM → Electromagnetic;
 R → Radiometric; GPR → Ground-Penetrating Radar.

Gravity Method

Introduction

- Gravity method is used where density contrasts are present in a geological structure, and the usual approach is to measure differences in gravity along a survey line at discrete interval.
- In gravity method, we are mostly interested in lateral variations in Earth structure, because these involve lateral variations in density.
- Gravity method cannot detect oil & gas directly, since oil & gas are of low density and if accumulated in a suitable trap, it can give a gravity low that can be detected by gravity method.
- Gravity surveys, conducted to search for oil & gas, are broad regional studies. One need to answer: is there a large and thick enough sedimentary basin to justify further exploration?
- Gravity method can answer this question inexpensively because sedimentary rocks have lower densities than basement rocks.
- Gravity method is only used for mineral exploration if substantial density contrasts are expected, e.g., chromite bodies have very high densities. Buried channels, which may contain gold or uranium, can be detected because they have relatively low density.

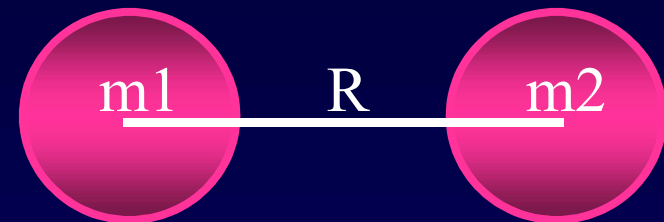
Density of common rocks and medium

Material	Density (gm/cm ³)
Air	~0
Water	1
Sea Water	1.03
Sediments	1.7-2.3
Sandstone	2.0-2.6
Shale	2.0-2.7
Limestone	2.5-2.8
Granite	2.5-2.8
Basalts	2.7-3.1
Metamorphic Rocks	2.6-3.0
Upper Mantle	3.3

Basic Theory in Gravity Method of Exploration

The fundamental equation used for mathematical treatment of the data and results in gravity method of exploration is Newton's Law of Gravitation:

$$F \propto \frac{M \times m}{R^2}$$



$$F = G \times \frac{M \times m}{R^2} \dots\dots\dots (1)$$

F: Gravitational force of attraction; M: Mass of the Earth, m: mass of object; R: separation distance, G is the Universal Gravitational Constant and $G = 6.6732 \times 10^{-8}$ cgs units or, $G = 6.6732 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$

The acceleration due to gravity can be expressed by combining Newton's Law of Gravitation with Newton's second law of motion.

The Global Gravity Field

- If the Earth were a perfect sphere with no lateral inhomogeneities and did not rotate, g would be the same everywhere and obey the formula:

$$g = GM / R^2$$

Newton's second law of motion:

$F = mg$ where m is the mass and g is the acceleration due to gravity. Solving equation (1) for the acceleration due to the Earth gives

$$g = Gm/R^2$$

- But this is not the case. We know that the Earth is inhomogeneous and it rotates.
- Rotation causes the Earth to be an oblate spheroid with an flattening of 1/298.257223563
- The polar radius of the Earth is ~21.385 km less than the equatorial radius, which means that g is ~0.67% less at equator than at poles.

Shape of the Earth and its effect on Gravity field

There are ways of specifying the dimensions of the spheroid.

By its major and minor radii a and b .

flattening $f = 1 - b/a$

eccentricity e ; $e^2 = 1 - b^2/a^2$

For the WGS84 spheroid,

$a = 6378.137$ km, $b = 6356.752$ km, and

$f = 1/298.257223563$

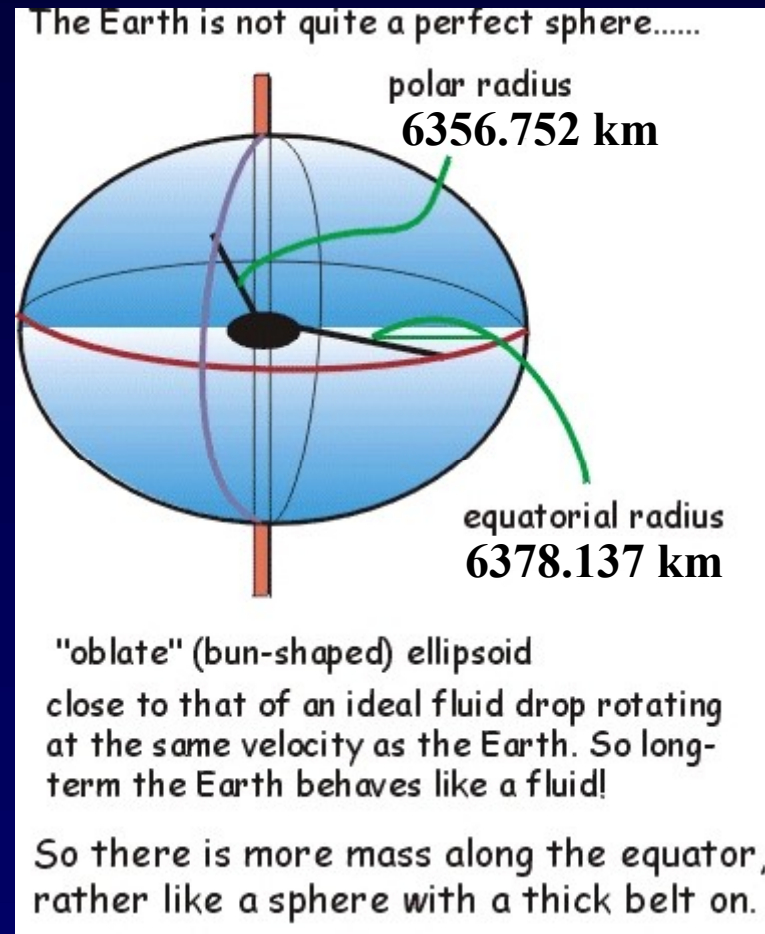
The eccentricity and flattening are thus related by:

$$e^2 = f(2-f)$$

Gravitational acceleration

At poles = **983.218 Gals**

At Equator = **978.049 Gals**



The Global Gravity Field

➤ The global gravity field

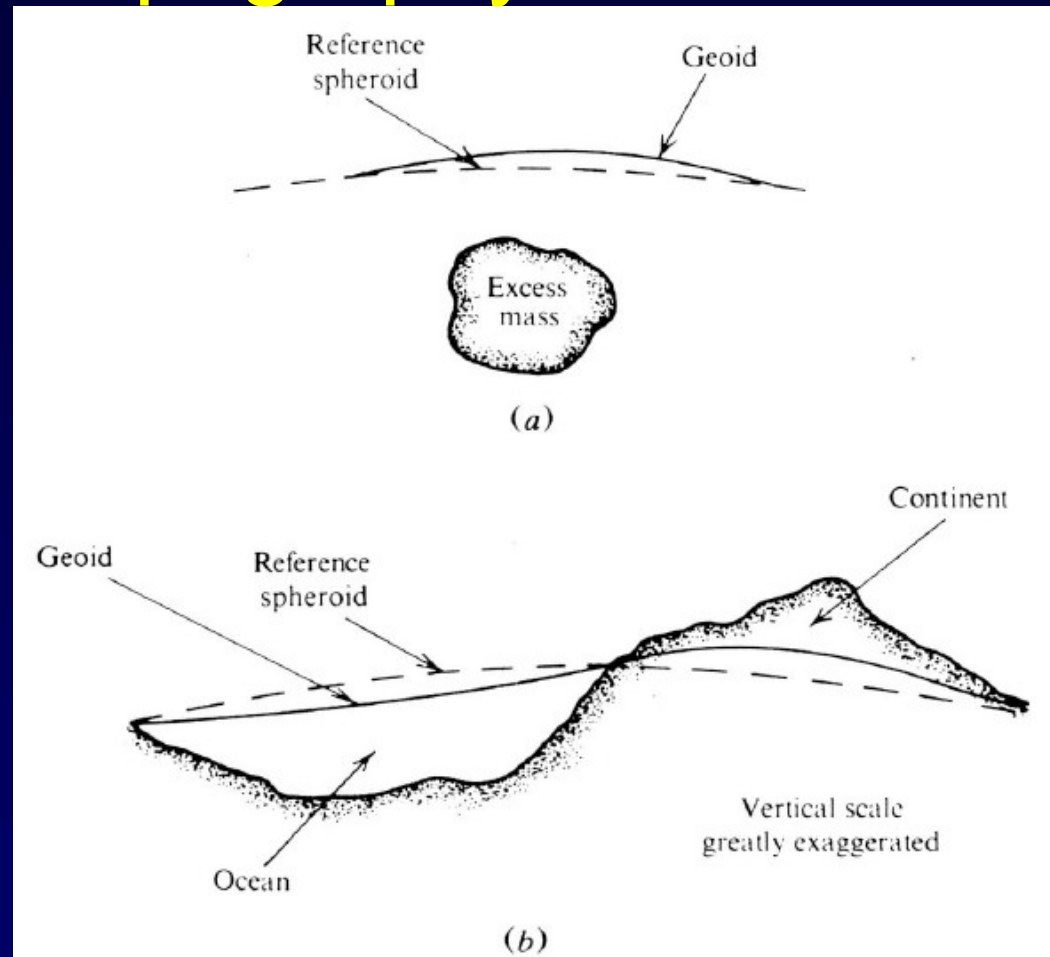
In geodesy, a reference ellipsoid is a mathematically defined surface that approximates the geoid. The best fitting spheroid (or ellipsoid which is a sphere flattened at the poles) is called the reference spheroid, and gravity on this surface is given by:

$$g(\theta) = 978031.846 (1 + 0.005278895 \sin^2\theta + 0.000023462 \sin^4\theta)$$

$g(\theta)$ is in mGal and the above formula is known as the International Gravity Formula 1967 based on the 1967 Geodetic Reference System (GRS67). θ is the geodetic latitude of the location.

Geoid is an equipotential surface corresponding to mean sea level. On land, it corresponds to the level that water would reach in canals connecting the seas.

The relationship between the geoid, the spheroid, topography and anomalous mass.



The geoid is a conceptual surface, which is warped due to absence or presence of attracting material. It is warped up on land and down at sea.

Unit of measurement of Gravity Anomaly

1 Gal (after Galileo) = 1 cm/s^2 .

g at the surface of the Earth is 980 Gal (cm/s^2) or $\sim 10^3$ Gals.

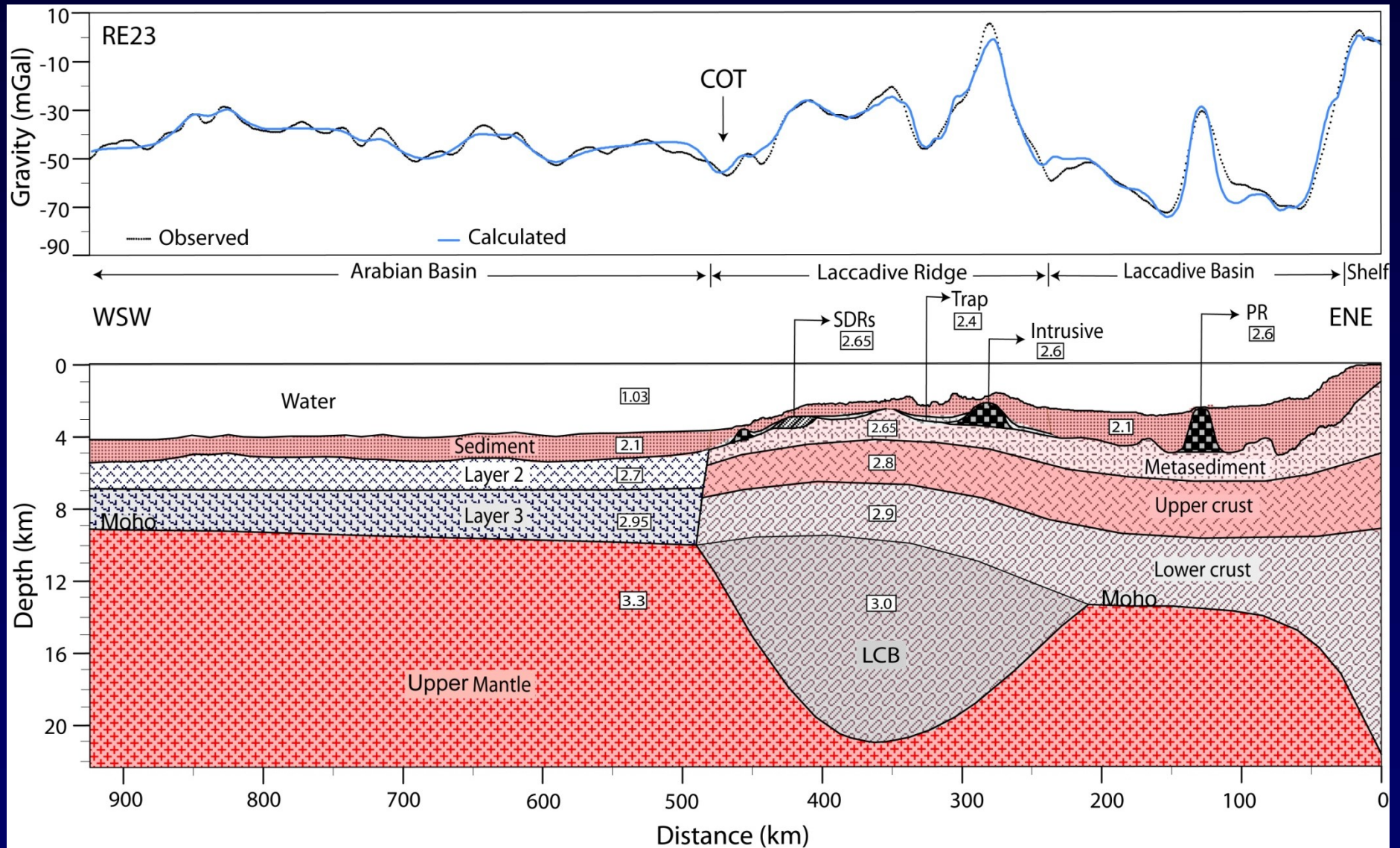
Gravity anomalies are measured in units of milliGals.

$1 \text{ mGal} = 10^{-3} \text{ Gals} = 10^{-3} \text{ cm/s}^2 = 10^{-3} \times 10^{-2} \text{ m/s}^2 = 10^{-5} \text{ m/s}^2$

$1 \text{ mGal} = 10^{-3} \text{ Gals} = 10^{-5} \text{ m/s}^2$, $\mu\text{Gal} = 10^{-6} \text{ Gals}$.

- Gravity meters, usually called gravimeters, are sensitive to 0.01 mGal which is equal to about 10^{-8} of the Earth's total value.
- Thus the specifications of gravimeters are amongst the most difficult to meet in any measuring device. It would be impossible to get the accuracy required in absolute gravity measurements quickly with any device, and thus field gravity surveying is done using relative gravimeters.

Crustal model from gravity anomaly data



COT: Continent Ocean Transition, Values in rectangular block is density in gm/cm³

Magnetic Method

Basic Principle of Magnetic Method

- The intensity of magnetization of rocks which determines its magnetic field, has both magnitude and direction and depends on the rock's susceptibility and permanent magnetization.
- In magnetic surveys the magnetic anomalies recorded in the field data are interpreted in terms of variation in magnetic susceptibility and /or remnant magnetism.
- Both of these rock properties can exist only at temperatures cooler than the Curie point, which limits the sources of magnetic anomalies to a maximum depth of 30 to 40 km.
- Magnetic method is the oldest method of geophysical exploration and is used to explore oil, minerals, and even archaeological artifacts.

Magnetic Force

Magnetic Force:

The expression of magnetic force is obtained from Coulomb's law for magnetic poles and symbolically is almost identical to Newton's law of gravity force. If two poles of strength P_o and P are separated by a distance r , the force F between them will be:

$$F = \frac{1}{\mu} \frac{P_o P}{r^2}$$

The constant μ is known as the permeability and depends upon the magnetic properties of the medium in which poles are situated. The value of μ (dimensionless) is precisely 1 in vacuum and is practically 1 in air.

Magnetic poles have either positive or negative polarity. If poles are of like type, the force is repulsive; if they are unlike, it is attractive. Magnetic poles of equal strength but opposite polarity always occur as a pair and are known as a magnetic dipole.

Strength of Magnetic Field / Magnetic Intensity / Magnetic Field Intensity

A more practical quantity than the force is the strength of magnetic field. The *Magnetic field strength* at a point is defined as the force per unit of pole strength which would be exerted upon a small pole of strength P_o if placed at that point. Thus the field strength H due to a pole of strength P at distance r away is given by:

$$H = \frac{F}{P_o} = \frac{P}{\mu r^2}$$

μ is the permeability, and $P_o \ll P$

Units of magnetic field intensity

In magnetic prospecting, variation in the intensity of Earth magnetic field is measured.

The unit of H is then expressed as lines of force per square of centimeters which in cgs system is designated as **1 dyne per unit pole** or as one **Oersted (Oe)**

The total magnetic field of the Earth is 0.5 Oe or 50,000 gamma.

$$1 \text{ gamma} = 10^{-5} \text{ Oe}$$

$$1 \text{ tesla} = 10^4 \text{ Oe}$$

Magnetic Susceptibility

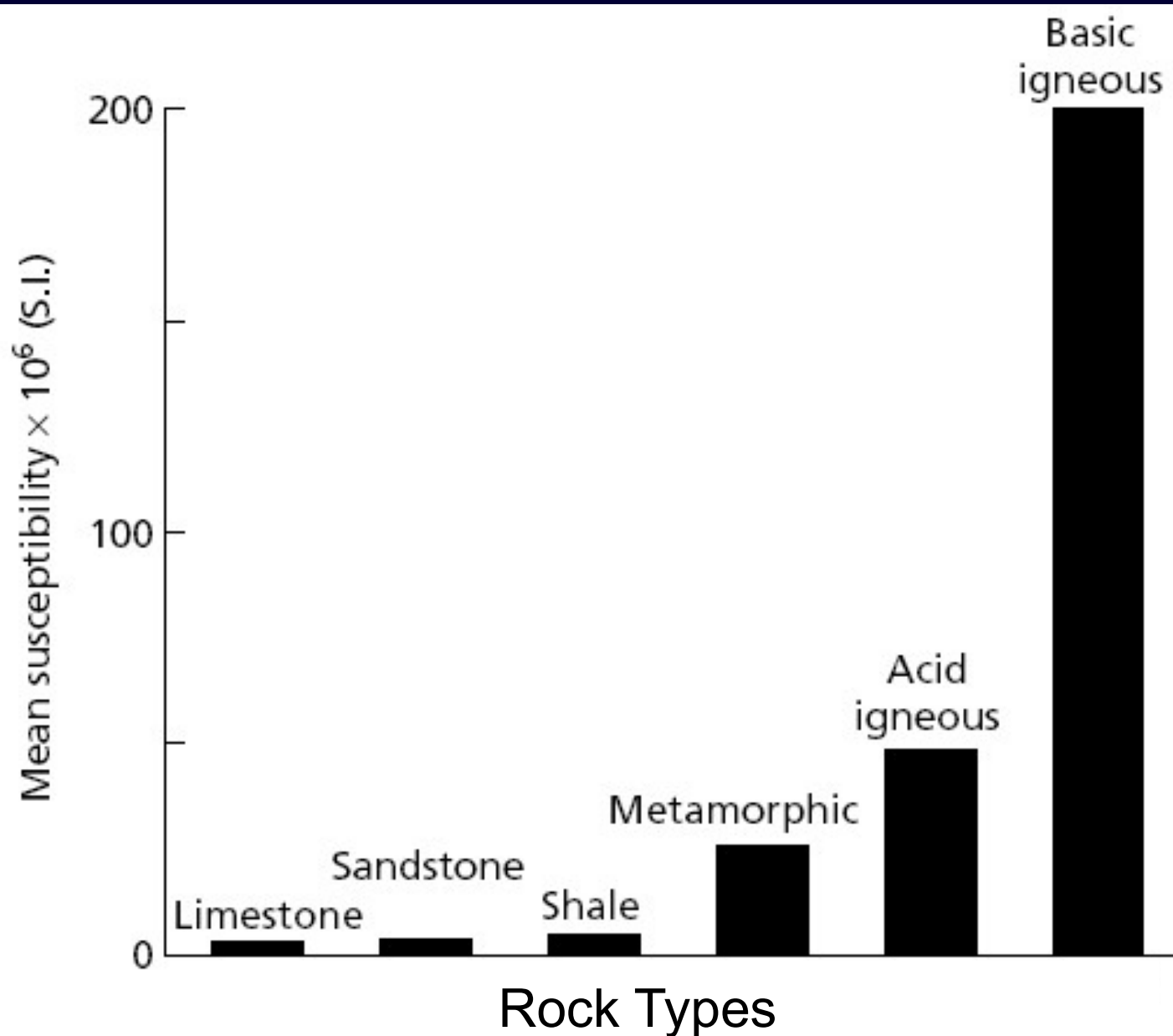
Magnetic characteristics of rocks depend largely on the content of magnetic minerals (such as magnetite) which are normally in the form of fine grains dispersed throughout the rock matrix.

When a magnetizable body is subjected to an external magnetizing field, H , it acquires a magnetization that is lost when the applied field is removed. Such a magnetization - induced by the applied field - is called as Induced Magnetization. For most rock bodies, the induced magnetization, J_i , is parallel and proportional to the applied field, H . This proportionality is valid for small field strengths and is expressed by the simple relationship:

$$J_i = kH$$

The factor k is called the magnetic susceptibility which is a dimensionless proportionality constant that indicates the degree of magnetization of a material in response to an applied magnetic field. In SI units, k is dimensionless. since J and H are both measured in the same units (A/m).

Magnetic Susceptibility of Common Rock Types



Remanent magnetization (Jr)

- Most rocks have a remanent (permanent) magnetization, Jr, in addition to Ji, induced by the Earth's field. Jr, may originate in different ways and at different times in the history of the rock. One principal type of Jr, especially found in igneous rocks, is Thermo-Remanent Magnetization (TRM) acquired by magnetic grains during cooling through the Curie temperature (about 580°C for magnetite grains) to normal atmospheric temperature in the presence of an external field (such as that of the earth).
- In general, Jr, is not related to the earth's present field but is governed instead by the field that existed when the rock was formed. The intensity of Jr, and its direction in a rock formation are measured on an oriented drill core in the laboratory using a spinner or astatic magnetometer.

Remanent magnetization...Contd

- Sedimentary rocks have typically low remanent intensities ($J_r < J_i$). The intensity of remanent magnetization (J_r) is particularly large in igneous and thermally metamorphosed rocks, often far exceeding J_i . Hence magnetic interpretation, especially in areas where such rocks occur, must take into account the influence of remanent magnetization on the magnetic anomaly.
- The same principle apply when object of investigations is iron or steel body

Total Magnetization

In general, the total magnetization, J , of a rock in-situ is expressed as the vector sum:

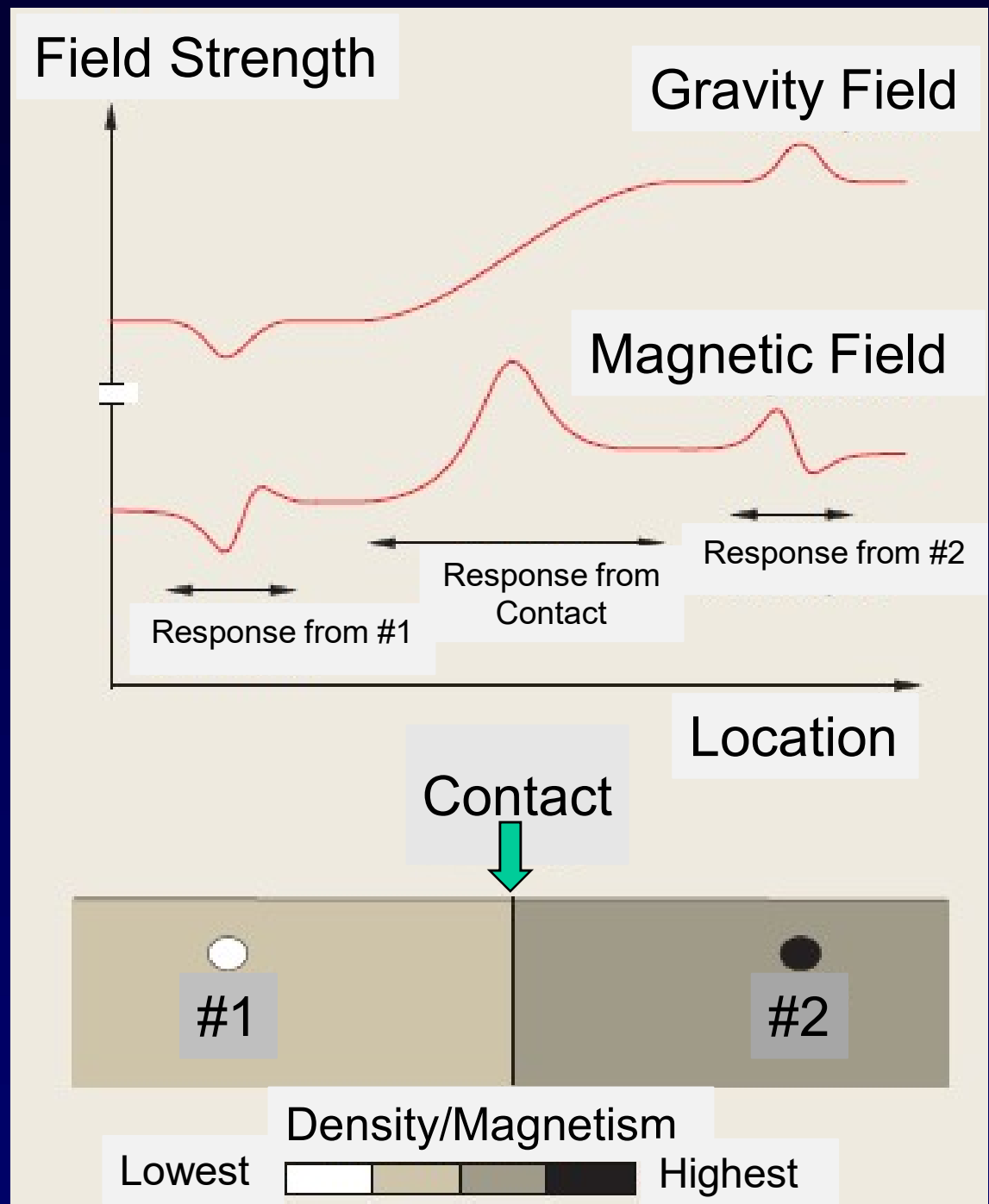
$$J = J_i + J_r$$

where J_i is in the direction of the earth's present field and J_r can have any arbitrary direction. The magnitude of the resultant vector, J , controls the amplitude of the magnetic anomaly caused by a rock body and the orientation of J influences anomaly shape. If J is constant and has the same direction throughout, a rock body is said to be uniformly magnetized.

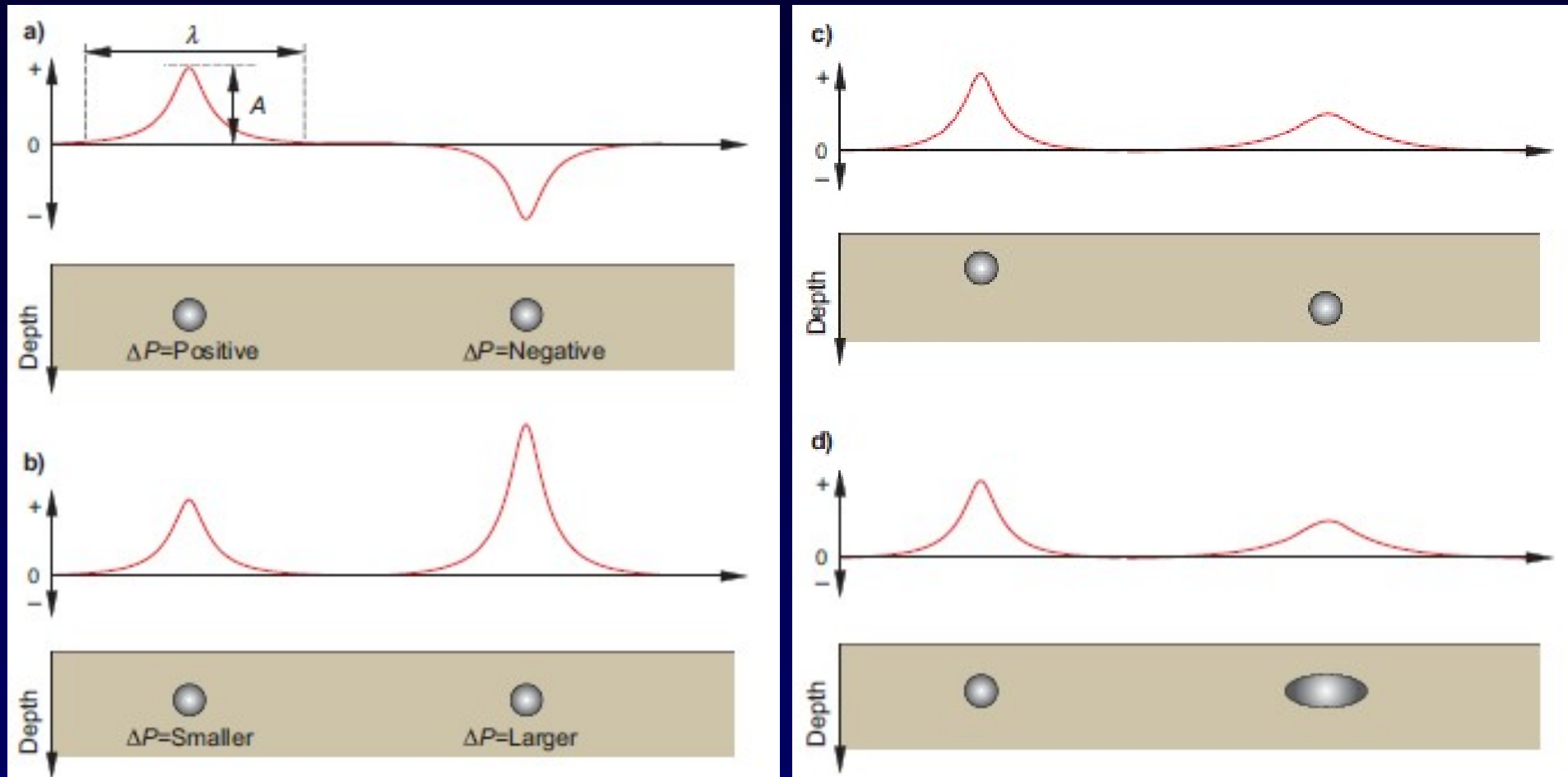
Applications of Magnetic Method

- Mapping depth to magnetic basement → Delineation of sedimentary basins useful for oil Exploration.
- Locating buried magnetized objects → (engineering and archaeological applications)
- Mineral exploration (iron rich mineral zones, placer deposits etc.)
- Subsurface structural features (faults, buried pipe lines etc.)
- Seafloor spreading anomalies (crustal ages – plate reconstructions)

Variations in the strength of the Earth's gravity and magnetic fields due to variations in crustal density and magnetism, respectively.



Geophysical responses of a simple shaped body



(a) The geophysical response due to contrast in physical property ($\Delta\rho$), (b) magnitude of the contrast, (c) depth of the source, and (d) shape of the source.

Similarities and differences between gravity and magnetic methods

Magnetic Method	Gravity Method
Passive and potential field method	Passive and potential field method
Mathematical expression for the force field is that of the inverse square law relation.	Mathematical expression for the force field is that of the inverse square law relation
Force between monopoles can either be attractive or repulsive	Force between masses is always attractive
A monopole cannot be isolated. Monopoles always exist in pairs (dipole).	A single point mass can be isolated.
A properly reduced field has variation due to variation in induced magnetization of susceptible rocks and remanent magnetization	A properly reduced field has variation due to density variation in rocks
Field changes significantly over time (secular variation).	Field does not change significantly over time.

Seismic Method

What is Seismic Method?

Seismic waves are directed into the earth. These waves pass through the earth and are reflected & refracted from boundaries of different geological horizons having enough seismic impedance (density * velocity) contrasts. The reflected and refracted energy is recorded by hydrophones/geophone placed at sea/land surface. The arrival times and amplitudes of these reflections are used to interpret the subsurface.

Why are the Seismic Methods Important

The seismic methods are important because it is useful to:

- map the interior of the earth.
- monitor the compliance of Comprehensive Test Ban Treaty (CTBT).
- detect contaminant of aquifer.
- find prospective oil and natural gas deposits.

Seismic Methods

```
graph TD; A[Seismic Methods] --> B[Reflection]; A --> C[Refraction]; B --> D[Useful to image subsurface layers]; C --> E[Useful to study large scale crustal layering (thickness and velocity)];
```

Reflection

Useful to image
subsurface layers

Refraction

Useful to study large
scale crustal layering
(thickness and velocity)

**Why do we prefer seismic technique
over other geophysical techniques?**

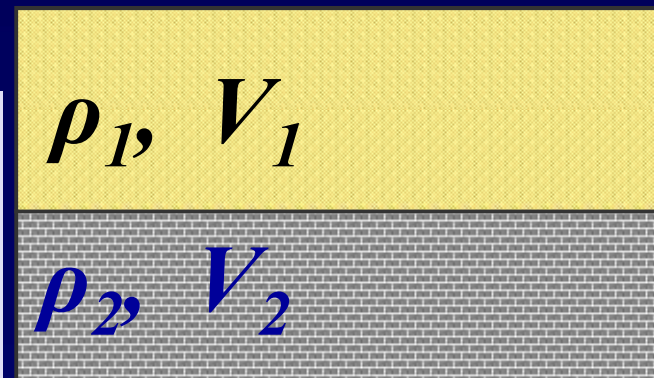
Why do we prefer seismic methods over other geophysical methods?

- The only method giving complete picture of the whole area.
- Give by far the best resolution among other geophysical methods.
- Map rock properties related to porosity and permeability, and presence of gas and fluids.
- Require significant logistics.
- Rely on extensive data processing and inversion.

Seismic Impedance

- Like any geophysical methods, seismic reflection imaging is also sensitive to only a specific physical property of rock.
- This property is called acoustic impedance: $I = \rho V$ (density x acoustic velocity).
- Seismic reflection amplitude is proportional to the relative *impedance contrast across a contact of two layers*. $R_{12} = A_i / A_r$:

$$R_{12} = \frac{I_2 - I_1}{I_2 + I_1} = \frac{\rho_2 V_2 - \rho_1 V_1}{\rho_2 V_2 + \rho_1 V_1}$$



R_{12} : Reflection Coefficient $\rightarrow -1 \leq R_{12} \leq 1$

Seismic Transmissivity

- The normal incidence transmissivity (T_{12}) can be defined as:

$$T_{12} = A_t / A_i$$

where A_t is the displacement amplitude of the transmitted wave. Continuity of displacement across the boundary requires:

$$R_{12} + T_{12} = 1$$

$$T_{12} = \frac{A_t}{A_i} = \frac{2I_1}{I_2 + I_1} = \frac{2\rho_1 V_1}{\rho_2 V_2 + \rho_1 V_1} \rightarrow 0 \leq T_{12} \leq 2$$

In practical terms, the equations for R_{12} & T_{12} describe the reflectivity and transmissivity only when seismic waves are normally incident (i.e. perpendicular) to the boundary. To describe the seismic reflectivity recorded with non zero offset (i.e. source and receivers in different locations), the above equations require modifications.

Significance of Impedance Contrast across a Contact of two layers

Please make a note of the values of reflection coefficient (R_{12}) and transmission coefficient (T_{12}) in following important special cases:

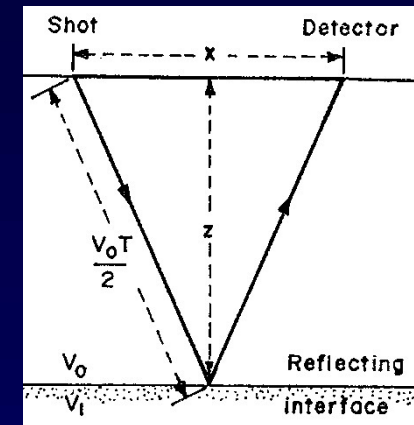
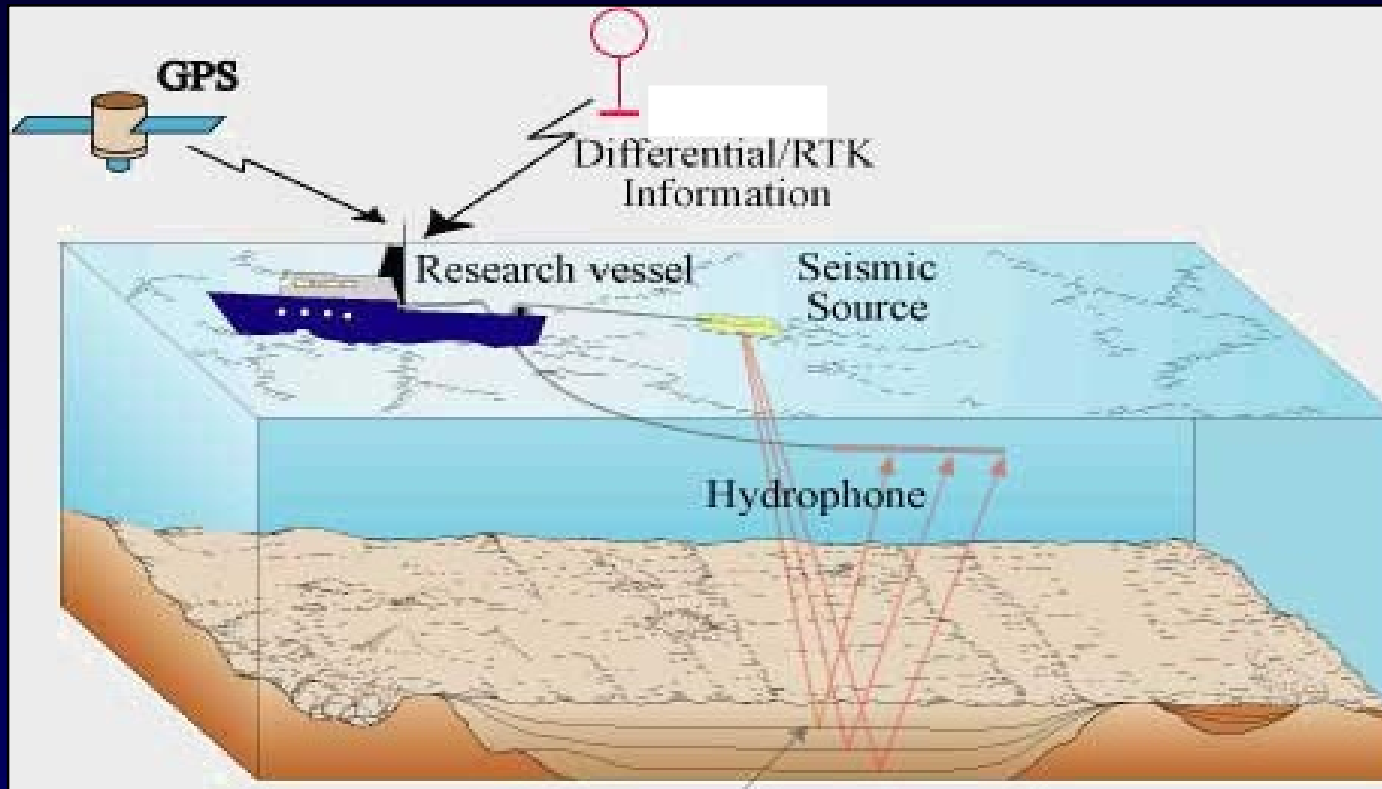
Case 1: If $I_1 = I_2 \rightarrow R = 0, T = 1$ (100% energy is transmitted)

Case 2: If $I_1 \gg I_2 \rightarrow R = -1, T = 2$.

The value $R = -1$ means that the pulse will be reflected with a polarity change, for example at the rock-air interface, with an upward traveling wave.

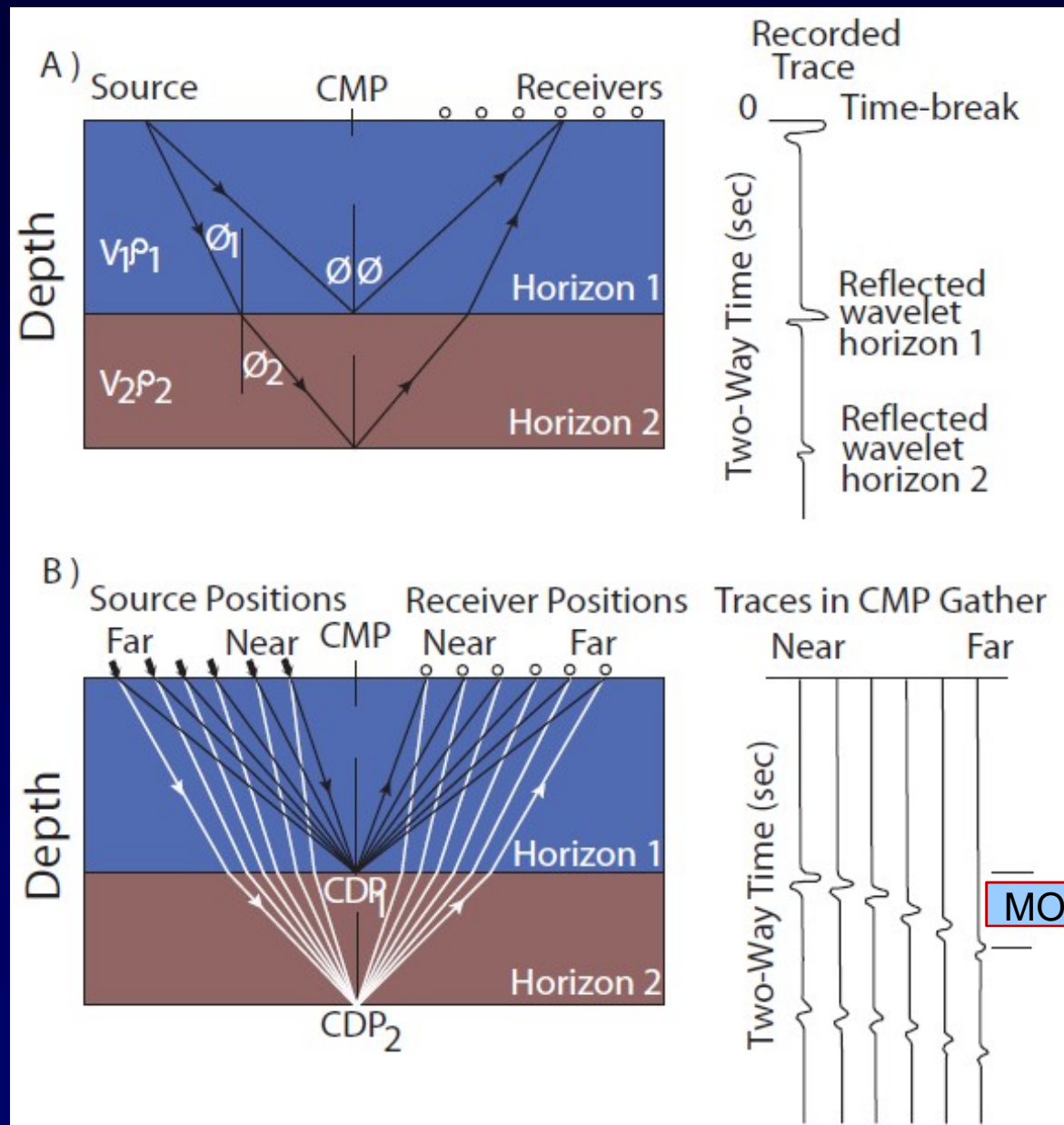
Case 3: If $I_2 \gg I_1 \rightarrow R = 1, T = 0$ (wave travelling down through the air and hitting the air-earth interface).

Principle of Seismic Reflection method



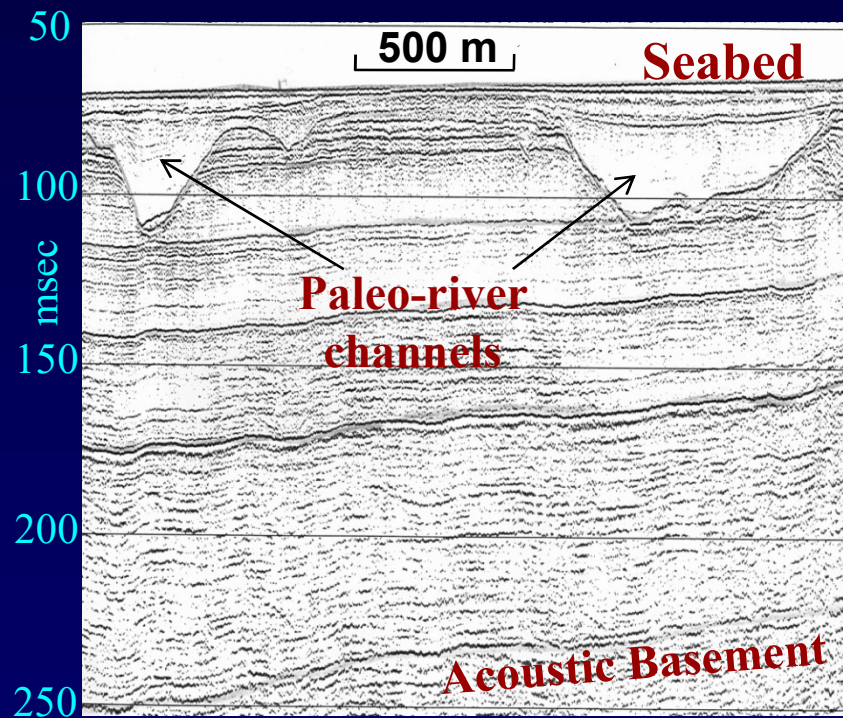
Seismic waves are directed into the earth. These waves pass through the earth and are reflected from boundaries of different geological horizons having enough seismic impedance (density * velocity) contrasts. The reflected energy is recorded by hydrophones/geophone placed at sea/land surface. The arrival times and amplitudes of these reflections are used to interpret the subsurface.

Reflection From Single Point of Sub-surface

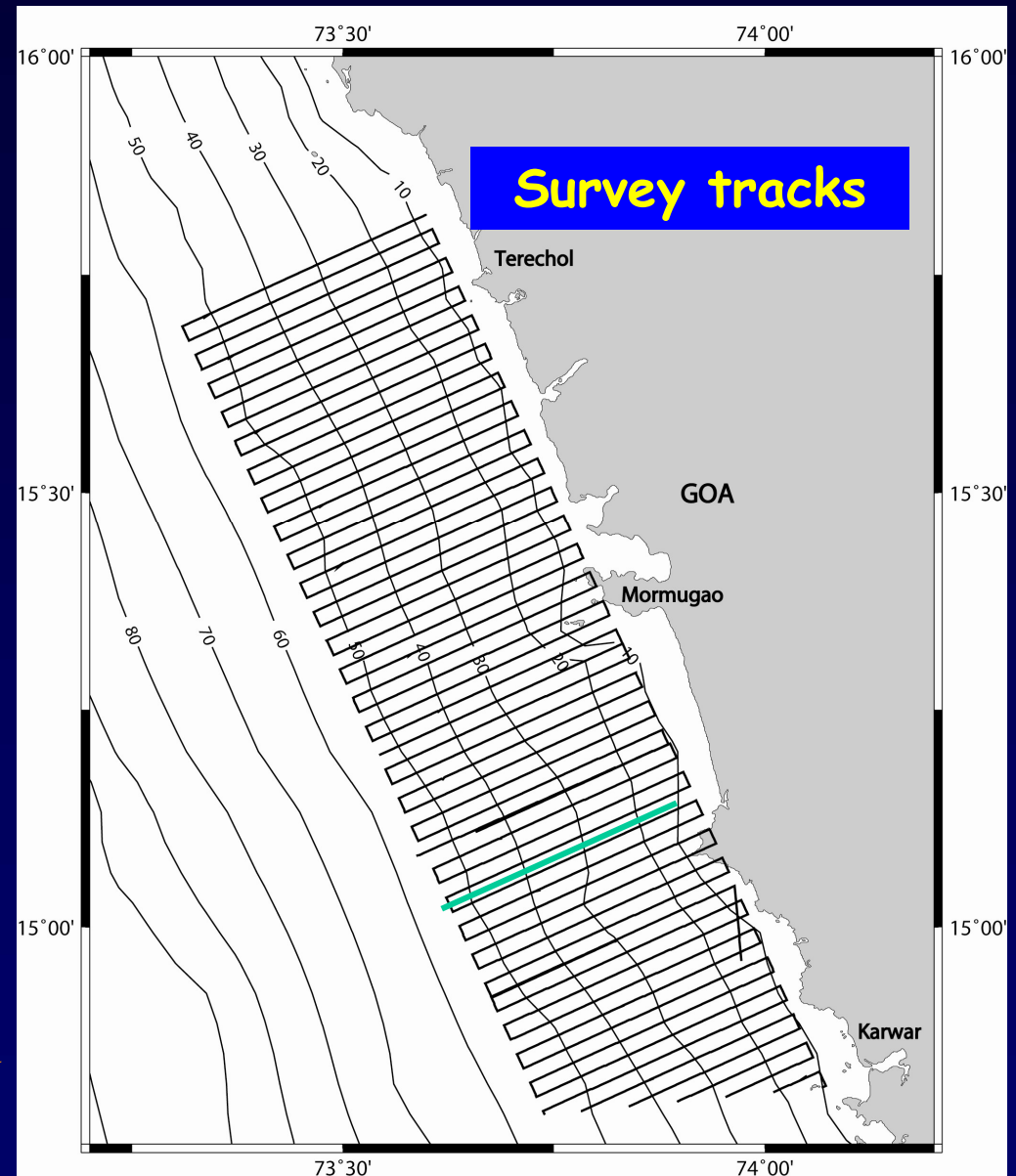


Seismic Section along Trackline

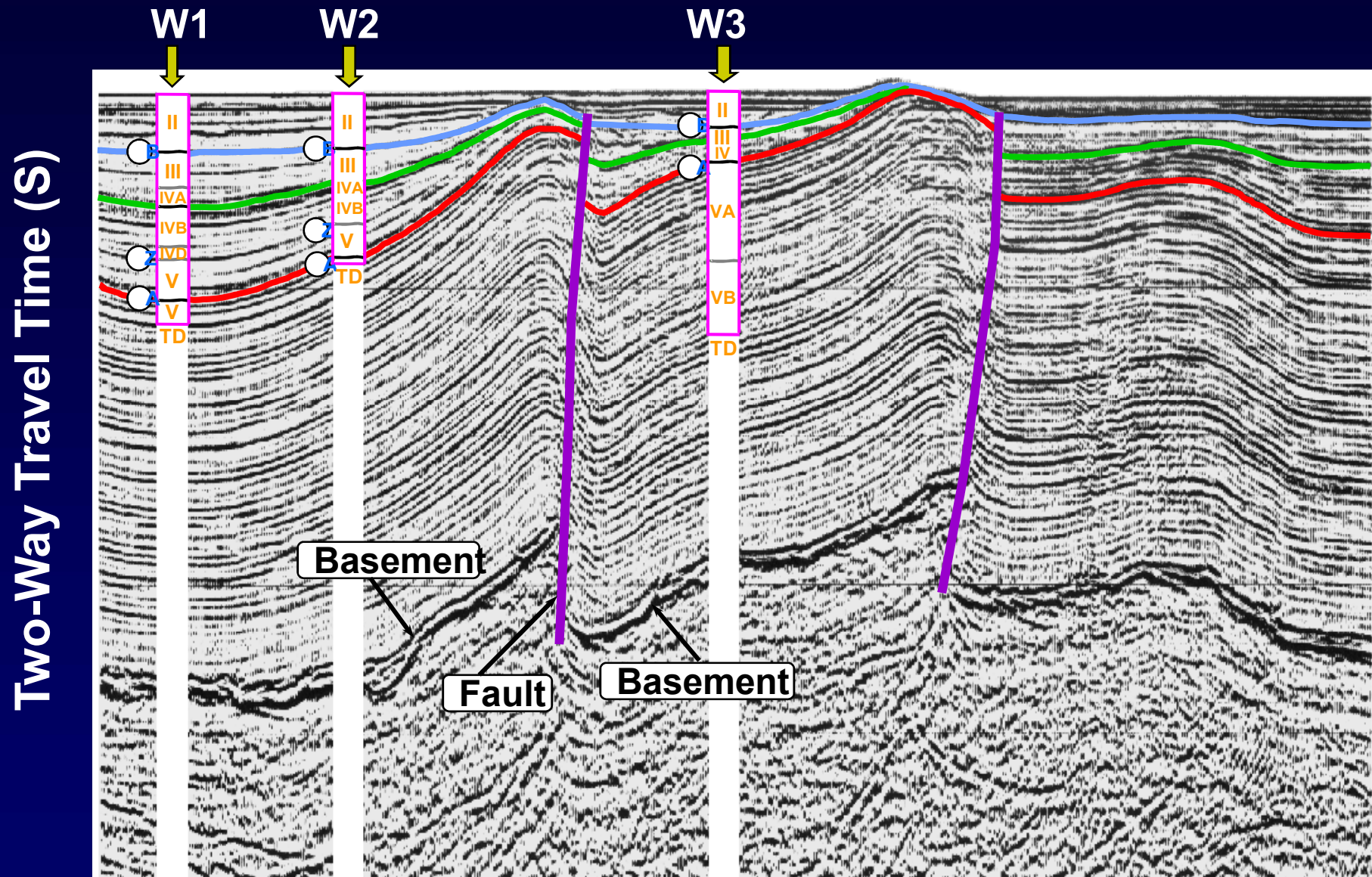
- Systematic high resolution geophysical exploration along coastal zone between 10 and 50 meters water depth indicated the presence of buried paleo-river channels.



High resolution shallow seismic record off Goa



Correlation of seismic reflector with drilled wells



Branches of Geophysics

Study of solid earth – Solid earth properties range from tectonic study to seismic analysis for earthquakes. This branch studies the oil and mineral deposits. The soils samples are analyzed for any unusual deposits or textures. Under this stream, the major branches are Petroleum geophysics Environmental Geophysics and Mining Geophysics.

Study of water – This is the study of fresh water as well as water under the earth surface. Study of water is done by hydrologists who analyze water cycles and water tables.

Oceanography - the study of the oceans and the ground beneath, also falls under this category.

Branches of Geophysics

Study of air – Air is a major component of the earth. The study of air helps in predicting weather conditions and guard against extreme conditions such as hurricanes. Under this stream the major branches are atmospheric geophysics, climatology

Life and geophysics – The interaction of organisms and the earth is an important factor. It is important to note that oil deposits mainly result from decaying matter. Conditions below the ground are studied to assess their effect on the existence of organisms.

Planetary geophysics - study of planets, celestial bodies and planetary systems and the processes of their formation using geophysical methods.

Branches of Geophysics

Biogeophysics – study of how plants, microbial activity and other organisms alter geologic materials and affect geophysical signatures.

Exploration geophysics – the use of surface methods to detect concentrations of ore minerals and hydrocarbons.

Geophysical fluid dynamics – study of naturally occurring, large-scale flows on Earth and other planets.

Geodesy – measurement and representation of the Earth, including its gravitational field.

Geodynamics – study of modes of transport deformation within the Earth: rock deformation, mantle convection, heat flow, and lithosphere dynamics.

Geomagnetism – study of the Earth's magnetic field, including its origin, telluric currents driven by the magnetic field, the Van Allen belts, and the interaction between the magnetosphere and the solar wind.

Branches of Geophysics

Mathematical geophysics – development and applications of mathematical methods and techniques for the solution of geophysical problems.

Mineral physics – science of materials that compose the interior of planets, particularly the Earth.

Near-surface geophysics – the use of geophysical methods to investigate small-scale features in the shallow (tens of meters) subsurface.

Paleomagnetism – measurement of the orientation of the Earth's magnetic field over the geologic past.

Seismology – study of the structure and composition of the Earth through seismic waves, and of surface deformations during earthquakes and seismic hazards.

Tectonophysics – study of the physical processes that cause and result from plate tectonics.

Geological Time Scale

- The geological time scale is the extensive interval of time occupied by the geologic history of Earth.
- It is used primarily by Earth scientists to describe the timing and relationships of events on a worldwide scale in geologic history of the Earth.
- The geologic time scale is a way of representing time based on events that have occurred throughout Earth's history, i.e. a time span of about 4.54 billion years.
- The geologic time scale grew out of necessity. No one person or expert committee proposed the geologic time scale used today. It grew by through the efforts of numerous geologists working independently. Today the recognition of formal subdivisions of geologic time is determined by international committees.

Geological Time Scale.....contd.

- It subdivides all time into named units of abstract time called - in descending order of duration - eons, eras, periods, epochs, and ages.
- An EON is the largest (formal) geochronologic time unit and contains the Archean, Proterozoic, and Phanerozoic. The Hadean is an informal EON, but is commonly used.
- An ERA is the second largest geochronologic time unit and as of now, there are currently ten defined eras.
- A PERIOD is a major rank below an era and above an epoch. As of now, there are currently 22 defined periods.
- An EPOCH is the second smallest geochronologic unit, between a period and an age. There are currently 37 defined and one informal epochs.

Geological Time Scale

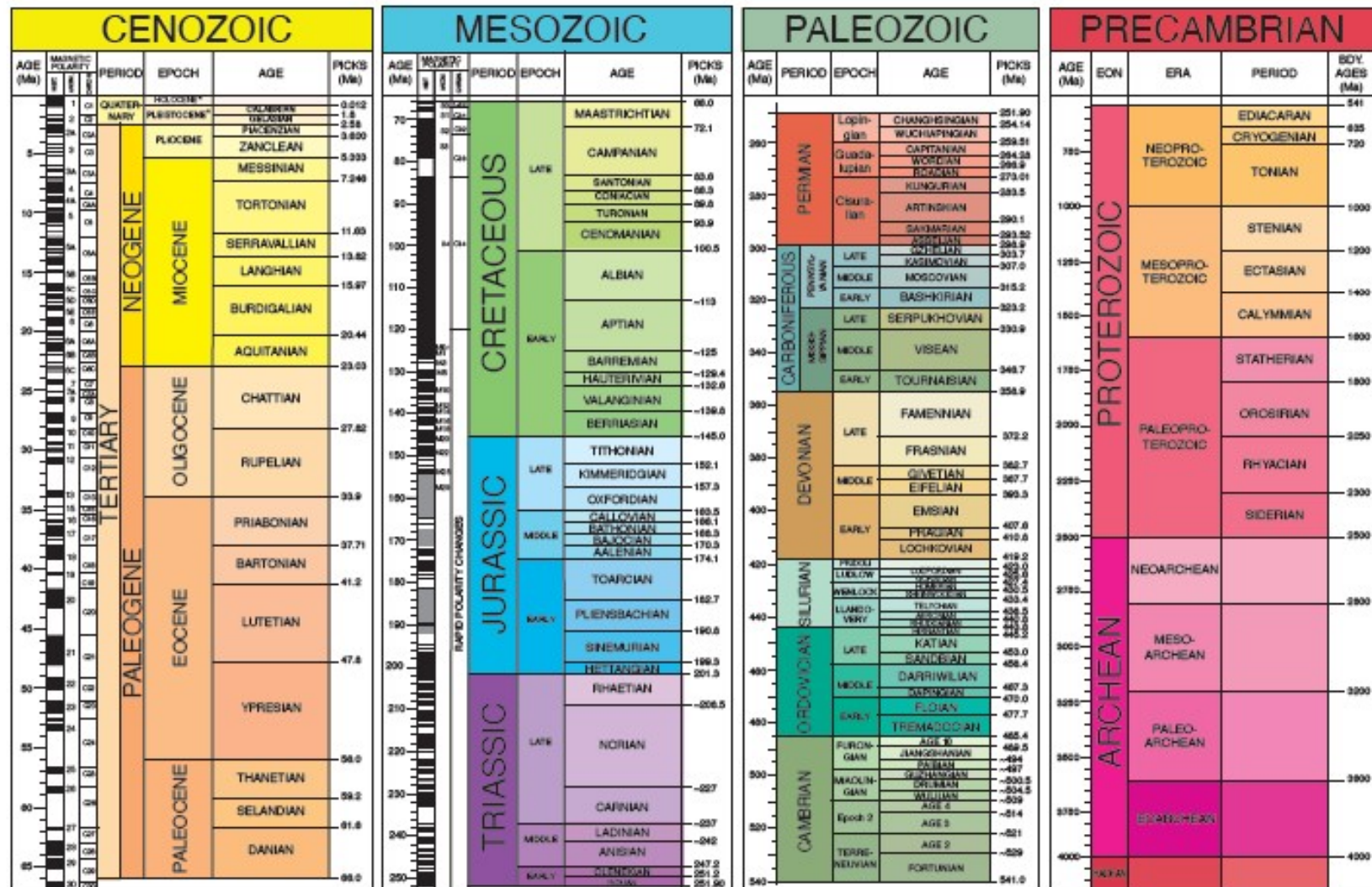
Eon	Era	Period	Epoch	MYA	Life Forms	North American Events
Phanerozoic	Cenozoic (CZ)	Quaternary (Q)	Holocene (H)	0.01	Age of Mammals	Ice age glaciations; glacial outburst floods
			Pleistocene (PE)			
		Tertiary (T)		2.6		Cascade volcanoes (W)
			Pliocene (PL)			Linking of North and South America (Isthmus of Panama)
			Miocene (MI)	5.3		Columbia River Basalt eruptions (NW)
				23.0		Basin and Range extension (W)
			Oligocene (OL)			
			Eocene (E)	33.9		
			Paleocene (EP)	56.0		Laramide Orogeny ends (W)
				66.0		Mass extinction
	Mesozoic (MZ)	Cretaceous (K)			Age of Reptiles	Laramide Orogeny (W)
				145.0		Western Interior Seaway (W)
		Jurassic (J)				Sevier Orogeny (W)
				201.3		Nevadan Orogeny (W)
		Triassic (TR)				Elko Orogeny (W)
	Paleozoic (PZ)	Permian (P)		251.9	Age of Amphibians	Mass extinction
		Pennsylvanian (PN)		298.9		Breakup of Pangaea begins
		Mississippian (M)		323.2		Sonoma Orogeny (W)
		Devonian (D)		358.9	Fishes	Supercontinent Pangaea intact
		Silurian (S)		419.2		Ouachita Orogeny (S)
		Ordovician (O)		443.8		Alleghany (Appalachian) Orogeny (E)
		Cambrian (C)		485.4	Marine Invertebrates	Ancestral Rocky Mountains (W)
				541.0		Antler Orogeny (W)
						Acadian Orogeny (E-NE)
Proterozoic	Archean	Precambrian (PC, W, X, Y, Z)				Taconic Orogeny (E-NE)
						Extensive oceans cover most of proto-North America (Laurentia)
Hadean						
				4600		Formation of the Earth

Geological Time Scalecontd.

- An age is the smallest hierarchical geochronologic unit. As of now, there are currently 96 formal and five informal ages.



GEOLOGIC TIME SCALE v. 6.0



Applications of Geophysical Methods

1. Hydrocarbon exploration
2. Regional geological studies
3. Exploration/development of mineral deposits
4. Engineering/environmental site investigation
5. Hydrogeological investigations
6. Detection of subsurface cavities
7. Mapping of contaminant plumes
8. Detection and location of buried metallic objects
9. Archaeogeophysics
10. Biogeophysics
11. Forensic geophysics (i.e., illegal burials, etc.)
12. Unexploded Ordnance (UXO detection and characterization)
13. Homeland Defence
14. Urban (utility mapping, underground storage tank location)

What are the benefits of Geophysical methods?.....contd.

The relevancy of the methods lies in the concrete and cost-effective benefits it delivers. These include:

1. **Non-destructive:** It is ideal for use in populated areas, such as cities, where many of today's environmental and engineering issues arise. It also means an archeological site can be examined without destroying it in the process.
2. **Efficiency:** It provides a means of evaluating large areas of the subsurface rapidly.
3. **Comprehensiveness:** Combinations of methods (i.e. multi-disciplinary methods) provide the means of applying different techniques to solve complex problems. The more physical properties that are evaluated, the less ambiguous the interpretation becomes.

What are the benefits of Geophysics?

.....Contd.

4. **Cost-effective:** Geophysics does not require excavation or direct access to subsurface (except in the case of borehole methods where access is typically by drilled holes). This means vast volumes of earth can be evaluated at far less cost than excavation or even grid-drilling methods.
5. **Proven:** The majority of techniques have been in existence for more than a half-century and are mature.



Thank You

Pillow lava at CIR observed during manned submersible dive